Visual Recognition using Deep Learning
2026 Spring, Homework 2

Release Date: 2026/03/31 15:40

Homework 2

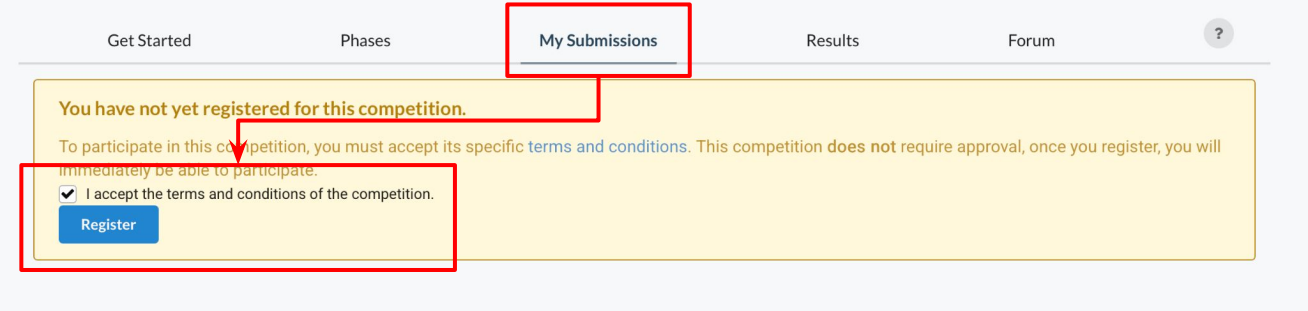
- Deadline: 23:59, 04/21, 2026 (Originally 04/14)
- **Participate the competition** (85%): Digit Detection Problem
  - Participant the competition on the CodaBench and get the highest score as possible. (80%)
  - Code reliability & quality (5%)
- **Report and code** (15%): Document your method and findings.
  - Report
    - In PDF format and written in English. (5pt penalty)
    - Introduction to your method (e.g., data pre-processing, model architecture, hyper-parameters)
    - Innovative ideas or additional experiments to further improve the model.
  - Code
    - Zip your code (.py) alone with report - Submit to E3.
    - You should also put your code on your GitHub repository and provide the link in the report.

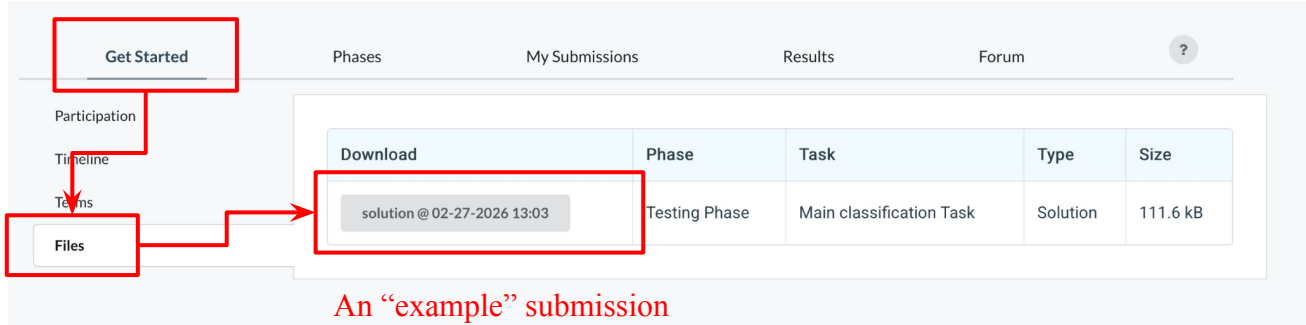
Links

- Link to the dataset
- Link to the competition

How to participate the competition and do submission

1. Register an account on CodaBench. (**Do not** use two accounts to participate in the competition.)
2. After you click the competition link, go to My Submissions, and join the competition.

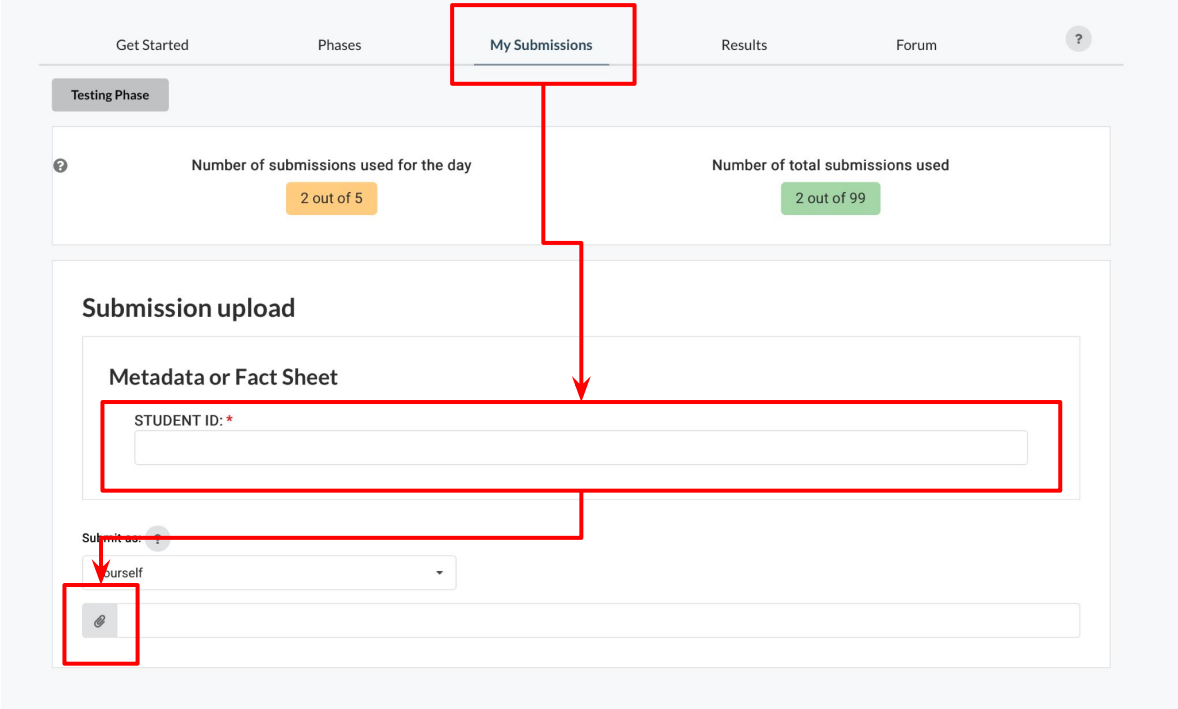




An "example" submission

How to participate the competition and do submission

3. Enter your student ID number, then upload the submission file.

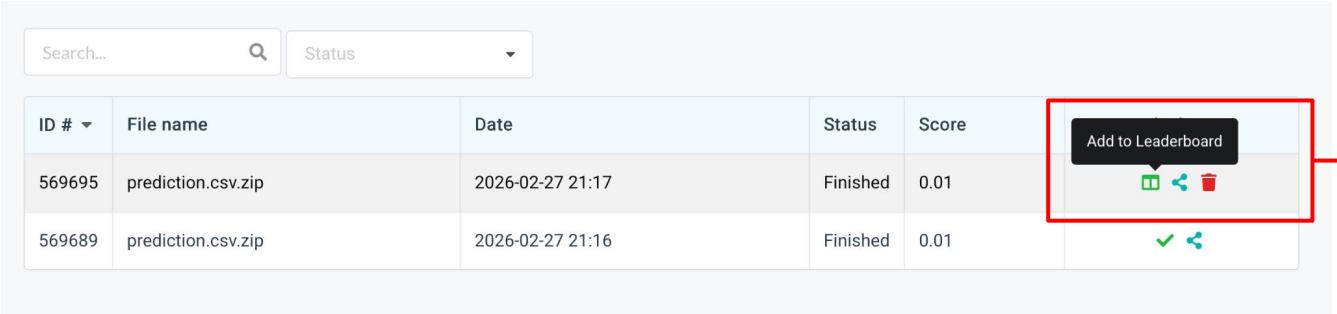


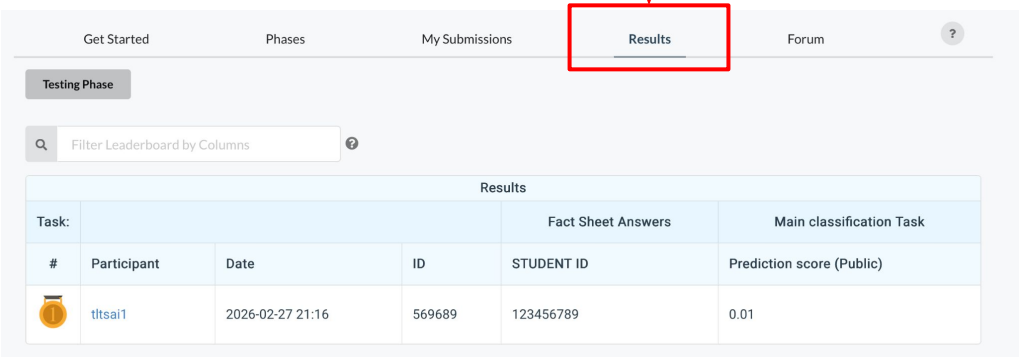
No limitation for the name of the "zip" file; however, inside the zip file, your **result file must be named "pred.json"**

How to participate the competition and do submission

4. Don't forget to "Add to Leaderboard".

5. Don't forget to check your results on the leaderboard.





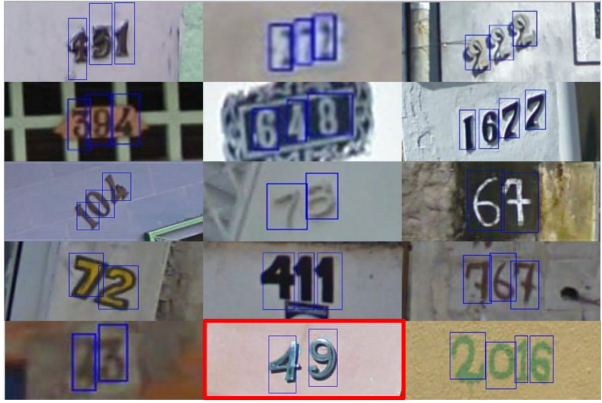
Coding Environment

- Recommnedation: Python 3.9 or higher
- Tips
  - We recommend you to use **virtual environments** when implementing your homework assignments.
  - Here are some popular virtual environment management tools
    - Poetry
    - Conda
    - Virtualenv

Numpy & PyTorch

- Numpy Tutorial: Link
- PyTorch Tutorial: Link
  - Free to use any modules and functions

Task and Dataset



- Task: Digit Detection
- Inputs
  - RGB images
    - Training / Validation: 30,062 / 3,340
    - Test: 13,068
- Target
  - The class and bounding box of each digit in the image, e.g., the bboxes of “4” and “9”.
- **Additional Requirements / Limitations** **(15 pts penalty, each)**
  - You must use DETR as the model for this task, with ResNet-50 as the backbone.
    - Other components of DETR (e.g., the encoder and decoder) may be modified.
  - Pretrained weights for the backbone are allowed.
  - No external data (i.e., data from other sources) is allowed.

Task and Dataset

- The training and validation labels are JSON files in COCO format.
  - The bounding boxes are described in the format [x_min, y_min, w, h] without normalization.
  - The category id starts from 1.
- The submission file should be a JSON file in COCO format named **pred.json**.
  - It should be a list of labels
  - Each label is represented as a dictionary with the keys: image_id, bbox, score, and category_id.


```
{
    "image_id": 2,
    "bbox": [
        98.08123016357422,
        14.60795783996582,
        10.317886352539062,
        20.47014808654785
    ],
    "score": 0.9570107460021973,
    "category_id": 3
},
{
    "image_id": 2,
    "bbox": [
        54.659828186035156,
        7.852010726928711,
        12.288619995117188,
        24.77677345275879
    ],
    "score": 0.7752974629402161,
    "category_id": 2
},
{
    "image_id": 2,
    "bbox": [
        90.57273864746094,
        15.257061958312988,
        9.276664733886719,
        19.51452922821045
    ],
    "score": 0.6907596588134766,
    "category_id": 2
},
```

Grading Policy - Report (15%)

- Format: PDF, written in English. (-5pts if not followed)
- Sections that you should include

12pts

- **Introduction** to the task and core idea of your method
- **Method:** Describe your data preprocessing procedure, model architecture, hyperparameter settings, and any modifications or adjustments you made.
- **Results**: your findings / model performance / visualization (e.g., training curve, confusion matrix, etc.)
- **References**: Your method references (paper / Github sources, must include if you use any). You may refer to the ECCV 2026 template for the citation and reference formatting guidelines.

We encourage you to stand on the shoulders of giants - only clone and run it is not enough.

3pts

- **Additional experiments** to explore better performance
  - Simply tuning the hyper-parameters doesn't count (e.g., batch-size, LR, different optimizers).
  - Hint: Try to add/remove some layers, use different loss functions, etc.
- You should 1) include your hypothesis (why you do this), 2) How this may (or may not) work, and 3) The experiment results and their implications.

Grading Policy - Code Reliability (5%)

1. Please follow the PEP8 instructions and lint your code.
2. Push your code to the GitHub
   - It should contain a README.md file that includes the following sections:
     - Introduction
     - Environment Setup
     - Usage
     - Performance Snapshot.

Python Coding Style Guide Reference

1. [PEP8]
2. [Google Python Style]

NYCU Computer Vision 2026 HW1

- **Student ID:** 123456789
- **Name:** Your Name

Introduction

Briefly introduce your work here.

Environment Setup

How to install dependencies.

```
pip install -r requirements.txt
```

Usage

Training

How to train your model.

```
python train.py
```

Inference

How to run inference.

```
python inference.py
```

Performance Snapshot

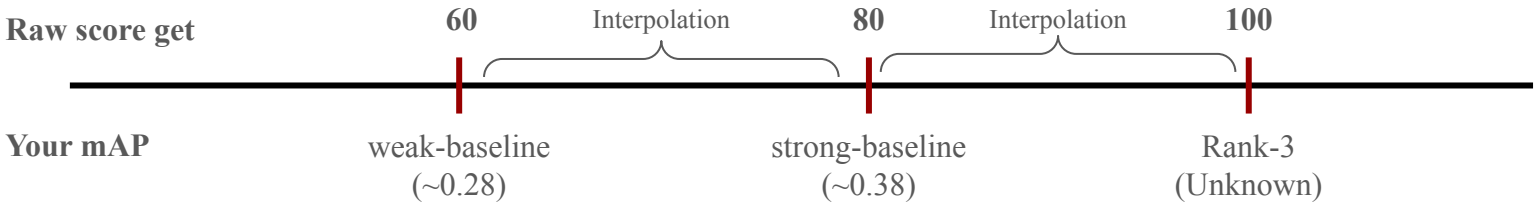
Insert a screenshot of the leaderboard here.

Grading Policy - Competition (80%)

We will use **private (hidden) leaderboard** to evaluate the performance (the distribution is similar for data in public and private set.) The public leaderboard is for you as reference.

Your score (competition):

- **Less than weak-baseline:** S = 0
- **Between weak-baseline and strong baseline:** 60 + (X - w.baseline) / (s.baseline - w.baseline) * (80 - 60)
- **Between strong-baseline and Rank3:** 80 + (X - s.baseline) / (rank3 - s.baseline) * (100 - 80)
- **Rank1, 2, 3:** 100



Submission

- Compress your **code** and **report** into a **.zip file** and submit it to E3.
  - Don't forget to push your code to GitHub. And your GitHub link should be written in the report.
- Report should be written in English.
- <STUDENT ID>_HW2.zip
  - codes (.py, folders, etc)
  - <STUDENT ID>_HW2.**pdf** (NO .doc, .docx or others format)
- Don't put the data (e.g. x.jpg / train.csv / test.csv) and model checkpoints into submission file. (-5pts if not followed)

Other rules

- **Late Policy**: A penalty of **20 points** per additional late day. (-20pts per delayed day)
  - For example, If you get 90 points but delay for two days, your will get only 50 points!

- **No Plagiarism**: You should complete the assignment by yourself. Students engaged in plagiarism will be penalized heavily. Super serious penalty.
  - e.g. -100pt for the assignment or failed this course, etc
  - Report to academic integrity office

FAQs

- Can I use any library/package/framework from GitHub or other resources?
  - **Yes**, we encourage you to learn how to leverage existing knowledge on your own task
    - e.g., Github of published works and model zoo from Torchvision
    - Focus on how to step forward from them - That's why part of scores comes from your competition ranks
  - You **should not copy-and-paste from your classmates** (Plagiarism)
- How to handle the GPU Out-of-Memory (OOM) issue?
  - Easy answer - Make your batch size smaller or make your model smaller.
  - Advanced methods: Try to figure it out by yourself. (Many online resources and AI-assistance)

FAQs

- If I don't have my own GPU – Use Google Colab
  - It should be 12 hours, please check this discussion in the stackoverflow
  - And some tricks here may make it longer.

- If you have other questions, ask on **E3 forum** first! We will reply as soon as possible.

It’s your turn! Have Fun!